Rafael D. De Paz
Systems Architect
me@rdepaz.com

March 24, 2026

Editorial Board
Foundations of Physics
Springer Nature

Dear Editorial Board,

I am writing to formally submit my manuscript entitled **"Occam-Decoherence: Parsimony as a Physical Thermodynamic Limit"** for peer-review in *Foundations of Physics*.

Quantum formulations famously struggle with wave-function probability cascades that expand geometrically, resulting in the "Many-Worlds" philosophical divide. However, when evaluating the universe exclusively as an execution grid bound by mathematical limits, I assert mathematically that absolute physical continuity does not allow infinite probability memory arrays.

This manuscript bridges physics with computer-science architecture, formally mapping Occam's Razor away from a subjective logical heuristic and cementing it as an explicit energy-saving algorithm inherent to local geometry grids. I demonstrate mathematically that physical arrays hit strict computation caps, forcibly triggering state reduction simply to conserve local processing integrity.

Because *Foundations of Physics* uniquely evaluates the conceptual frameworks surrounding the limits of established mechanics, we find this paper seamlessly aligns with your philosophical rigor and strict quantitative bounds.

I look forward to the insights of the reviewers.

Sincerely,

Rafael D. De Paz